

Randy R. Davila

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Education

University of Johannesburg <i>PhD in Mathematics</i> ○ Dissertation: <i>Total and Zero Forcing in Graphs</i> . Link ○ Advisor: Michael A. Henning, PhD	Johannesburg, South Africa 2016–2019
Rice University <i>M.A. in Computational and Applied Mathematics</i> ○ Thesis: <i>Bounding the Forcing Number of Graphs</i> . Link ○ Advisor: Illya V. Hicks, PhD	Houston, TX, United States 2013–2015
University of Houston–Downtown <i>B.S. in Applied Mathematics</i> ○ Minor: Applied Statistics	Houston, TX, United States 2009–2012

Appointments and Experience

Current Positions	
Engineering & Research <i>Member of Technical Staff (Engineering), Toronto, Canada</i>	FirstPrinciples 2025–Present
Department of Computational Applied Mathematics & Operations Research <i>Lecturer, Houston, TX</i>	Rice University 2021–Present
Previous Academic and Industry Positions	
Compiler Team <i>Library & Software Engineer, Berkeley, CA, United States</i>	RelationalAI 2022–2025
Department of Mathematics and Statistics <i>Assistant Professor of Data Science, Houston, TX</i>	University of Houston–Downtown 2019–2022
Department of Computer Science & Engineering Technology <i>Adjunct Professor of Computer Science, Houston, TX</i>	University of Houston–Downtown Spring 2020
Department of Mathematics and Statistics <i>Lecturer, Houston, TX</i>	University of Houston–Downtown 2017–2019
Department of Mathematics <i>Lecturer, San Marcos, TX</i>	Texas State University 2015–2017
Department of Computational & Applied Mathematics <i>Research Assistant, Houston, TX</i>	Rice University 2015
Department of Mathematics <i>Graduate Teaching Assistant, San Marcos, TX</i>	Texas State University 2012–2013

Research Interests

AI for Mathematical Discovery and Scientific Reasoning: Automated conjecture generation and ranking;

counterexample search and theory revision; optimization- and learning-based conjecturing pipelines; neural and transformer architectures for mathematical reasoning; AI systems for discovery in graph theory, number theory, and quantum information.

Graph Theory and Combinatorial Optimization: Extremal and structural results involving independence, domination and its variants (total, Roman, restrained), and zero forcing parameters (standard, positive semidefinite, power variants). Integer programming and algorithmic methods for NP-hard graph invariants; applications to network resilience, contagion processes, and relational systems.

Research Papers

Preprints

- *Impurity Graphs, Duality, and Real-Rooted Hamiltonians in $Y^{p,q}$* (with Y-H. Hu and S. Franco), to be submitted 2026.
- *From Counterexamples to Theorems: The THEO-CONJECTURE Loop in Scientific Discovery*, (with Y-H. Hu and E. Pitchford), submitted 2026.
- *Prime-Transparent Intervals and the Residue of Common-Divisor Graphs*, submitted 2026.
- *Pressure Monotonicity, Jacobi Curvature, and Fragmentation of Capillary Branches*, submitted 2026.
- *Neighbor Nominations and Total Domination in Graphs* (with M. A. Henning), submitted 2026.
- *Bond-Imbalance Participation in Benzenoid Hydrocarbons: Product Laws, Sharp Families, and Hückel π -Electron Energy*, submitted 2026.
- *Skew Forts and Cycle Structure in Claw-Free Graphs*, submitted 2026.
- *Local Forcing Pieces and Polynomial Upper Bounds for the Zero Forcing Number*, submitted 2026.
- *Exact Fidelity–Unitarity Frontiers for Unital Qubit Channels*, submitted 2026.
- *Sharp Pairwise-to-Global Joint-Measurability Ratios for Unbiased Qubit Measurements*, submitted 2026.
- *Cops and zombies in claw-free graphs* (with M. A. Henning), submitted 2026.
- *Havel–Hakimi Residues of Common-Divisor Graphs: Complete Asymptotics and Prime Spectral Filters*, to be submitted 2026. **ArXiv**
- *Geometry-Only CSL/DP Ratios and the Nonuniqueness of Decoherence Kernels* (with G. J. Milburn), submitted 2026. **ArXiv**
- *The zombie damage number of a graphs*, submitted 2026. **ArXiv**
- *Zero forcing in rooted product graphs and zero forcing density* (with M. Arulperumjothi, M. Subha, and S. Prabhu), submitted 2026.
- *Dual-server independent domination in graphs* (with M. Chellali, T. W. Haynes, S. T. Hedetniemi, and R. Pepper), submitted 2026.
- *Another conjecture of TxGraffiti concerning zero forcing and domination in graphs*, submitted 2026. **ArXiv**
- *On connected zero forcing sets: Density, enumeration, and products* (with B. Brimkov and H. Schuerger), submitted 2025. **ArXiv**
- *A characterization of claw-free graphs using zero forcing invariants* (with H. Schuerger and B. Small), *Journal of Graph Theory*, first-round revisions submitted in September 2025. **ArXiv**
- *Executable Mathematical Artifacts in Discovery: An Advisor-Supervised AI Loop for Conjectures, Counterexamples, and Proofs*, *Journal of Automated Reasoning*, first-round revisions submitted 2026.

Publications

- *On conjectures involving the dual-server independent domination number* (with M. Chellali, T.W. Haynes, S.T. Hedetniemi, and R. Pepper), *Journal of Combinatorial Mathematics and Combinatorial Computing*, to appear 2026.
- *Graffiti3Loop: Generating and Resolving Formal Machine-Generated Graph Conjectures in a Loop* (with M. R. Douglas, P. Vonderlind, W. Hu, and S. Frieder), *3rd AI for Math Workshop: Toward Self-Evolving Scientific Agents at the 43rd International Conference on Machine Learning (ICML)*, Canada, 2026. **Link**
- *Independence, induced subgraphs, and domination in $K_{1,r}$ -free graphs* (with Y. Caro, M. A. Henning, and R. Pepper), *Discrete Mathematics*, 2026. **ArXiv**

- *Automated conjecturing with TxGraffiti*, *Annals of Mathematics and Artificial Intelligence*, 2026. [Link](#)
- GAMBIT: Generating Automated Mathematical Bounds, Inequalities, and Theorems, *The 5th Workshop on Mathematical Reasoning and AI at NeurIPS 2025*, (2025). [Link](#)
- GraphCalc: A Python Package for Computing Graph Invariants in Automated Conjecturing Systems, *Journal of Open Source Software*, **10**(112), 8383, 2025. [Link](#)
- Inequality ranking and inference system (IRIS): Giving mathematical conjectures numerical value (with J. A. De Loera, J. Eddy, E. Fang, J. Lu, and Z. Yang), *2nd AI for Math Workshop @ ICML*, 2025. [Link](#)
- Lower bounds on the total distance k -domination number of graphs, *Theory and Applications of Graphs*, Vol. 12: Iss. 1, Article 6, 2025. [Link](#)
- Comparing the p -independence number of regular graphs to the q -independence number of their line graphs (with Y. Caro and R. Pepper), *Discrete Applied Mathematics*, **373**(2025), 316–326. [Link](#)
- On a conjecture of TxGraffiti: Relating zero forcing and vertex covers in graphs (with B. Brimkov, H. Schuerger, and M. Young), *Discrete Applied Mathematics*, **359**(2024), 290–302. [Link](#)
- Zero and total forcing dense graphs (with M. A. Henning and R. Pepper), *Discussiones Mathematicae Graph Theory*, **43** (2023), 619–634. [Link](#)
- The contagion number: How fast can disease spread? (with M. Blessley, T. Hale, and R. Pepper), *Decision Making: Applications in Management and Engineering*, **6**(1) (2023). [Link](#)
- Conjectures of TxGraffiti: Independence, domination, and matchings (with Y. Caro, M. A. Henning, and R. Pepper), *Australasian Journal of Combinatorics*, **84**(2) (2022), 258–274. [Link](#)
- New results relating independence and matchings (with Y. Caro and R. Pepper), *Discussiones Mathematicae Graph Theory*, **42**(3) (2022), 921–935. [Link](#)
- Relating zero forcing and domination in cubic graphs (with M. A. Henning), *Journal of Combinatorial Optimization*, **41** (2021), 553–577. [Link](#)
- Total forcing sets and zero forcing sets in trees (with M. A. Henning), *Discussiones Mathematicae Graph Theory*, **40** (2020), 733–754. [Link](#)
- On a Vizing-type integer domination conjecture (with E. Krop), *Theory and Applications of Graphs*, Volume 7: Iss. 1, Article 4, 2020. [Link](#)
- Zero forcing in claw-free cubic graphs (with M. A. Henning), *Bulletin of the Malaysian Mathematical Sciences Society*, **43** (2020), 673–688. [Link](#)
- The sub- k -domination number of a graph with applications to k -domination (with D. Amos, J. Asplund, and B. Brimkov), *Discussiones Mathematicae Graph Theory*, **40** (2020), 209–225. [Link](#)
- Extremal k -forcing sets in oriented graphs (with Y. Caro and R. Pepper), *Discrete Applied Mathematics*, **262** (2019), 42–55. [Link](#)
- Total forcing versus total domination in cubic graphs (with M. A. Henning), *Applied Mathematics and Computation*, **354** (2019), 385–395. [Link](#)
- A Vizing-type result for semi-total domination (with J. Asplund and E. Krop), *Discrete Applied Mathematics*, **258** (2019), 8–12. [Link](#)
- On the total forcing number of a graph (with M. A. Henning), *Discrete Applied Mathematics*, **257** (2019), 115–127. [Link](#)
- Matching, path covers, and total forcing sets (with M. A. Henning), *Quaestiones Mathematicae*, **43** (2019), 131–147. [Link](#)
- Bounds on the connected forcing number of a graph (with M. A. Henning, C. Magnant, and R. Pepper), *Graphs and Combinatorics*, **34** (2018), 1159–1174. [Link](#)
- A lower bound on the zero forcing number of a graph (with T. Malinowski and S. Stephen), *Discrete Applied Mathematics*, **250** (2018), 363–367. [Link](#)
- Total forcing and zero forcing in cubic claw-free graphs (with M. A. Henning), *Graphs and Combinatorics*, **34** (2018), 1371–1384. [Link](#)
- The forcing number of graphs with a given girth (with M. A. Henning), *Quaestiones Mathematicae*, **41**(2) (2018). [Link](#)
- Lower bounds on the distance domination number of a graph (with C. Fast, M. A. Henning, and F. Kenter), *Contributions to Discrete Mathematics*, **12**(2) (2017), 1–21. [Link](#)

- Bounds for the zero forcing number of graphs with large girth (with F. Kenter), *Theory and Applications of Graphs*, Volume 2: Iss. 2, Article 1, 2015. [Link](#)
- Upper bounds on the k -forcing number of a graph (with D. Amos, Y. Caro, and R. Pepper), *Discrete Applied Mathematics*, **181** (2015), 1–10. [Link](#)
- On the k -residue of disjoint unions with applications to k -independence (with D. Amos and R. Pepper), *Discrete Mathematics*, **321** (2014), 24–34. [Link](#)

Software

- *TxGraffiti* – Python package for automated mathematical and scientific discovery, 2025. GitHub: [RandyRDavila/TxGraffiti2](#).
- *GraphCalc* – Python package for computation of graph invariants and properties, 2024. GitHub: [RandyRDavila/graphcalc](#).
- *GraphCommunities.jl* – Julia package for community detection, 2023. GitHub: [RandyRDavila/GraphCommunities.jl](#).
- *GrinPy* – Extension of NetworkX for calculating graph invariants (with D. Amos), 2019. PyPI / GitHub: [grinpy](#).

Grants and Funding

2021–2022: Navy ROTC Cybersecurity Training Program

Sponsor: Office of Naval Research

Title: “CyberCops: The Houston Alliance for Student Research on Deceptive Attacks.”

Total amount: \$250,000. Collaborative grant between University of Houston (PI: Rakesh Verma), Texas Southern University (Co-PI: Ilija Jegdic), and University of Houston–Downtown (Co-PIs: Randy Davila and Katarina Jegdic).

Invited Presentations

2026:

- Automated Conjecturing, Internal Theories, and AI Discovery, Department of Mathematics Colloquium, Tarleton State University, Stephenville, TX.
- Automated Conjecturing, Internal Theories, and AI Discovery, Aarhus Mathematics & AI Workshop, Aarhus Institute of Advanced Studies (AIAS), Aarhus, Denmark.

2025:

- *Graffiti3*: Toward a Unified System for AI Discovery in Mathematics, Department of Computational Applied Mathematics & Operations Research Colloquium, Rice University, Houston, TX.
- Automated Conjecturing and Discovery in Mathematics, Mathematics & Machine Learning Seminar, California Institute of Technology, Pasadena, CA.
- Recent Developments in Automated Conjecturing, Math and Machine Learning Reunion Workshop, Harvard University, Cambridge, MA.
- Discovery in Mathematics with Automated Conjecturing, Special Session on Combinatorics and Computers, MAA MathFest, Sacramento, CA.
- Discovery in Mathematics with Automated Conjecturing, CMSA New Technologies in Mathematics Seminar, Harvard University, Cambridge, MA.
- Artificial Intelligence and Machine Learning Conjecturing with *TxGraffiti*, AMS Special Session on Machine Learning to Accelerate Mathematical Discovery (II), Joint Mathematics Meetings, Seattle, WA.

2023: Programming Computers to Behave like Mathematicians, Department of Mathematics & Statistics Colloquium, University of Houston–Downtown, Houston, TX.

2020: Data Driven Automated Conjecturing with *TxGraffiti*, Discrete Mathematics Seminar, Texas State University, San Marcos, TX.

2018: UHD Helped Transform My Life, College of Science and Technology Freshman Welcome Week Presentation, University of Houston–Downtown, Houston, TX.

2017:

- Conjecturing with *TxGraffiti*, Computational and Applied Mathematics Graduate Seminar, Rice University, Houston, TX.
- *TxGraffiti*, CombinaTexas, Texas A&M University, College Station, TX.
- Binding Independence and Domination with Annihilation and Sub-Domination, Discrete Mathematics Seminar, Texas State University, San Marcos, TX.
- Bounds on the Connected Forcing Number of a Graph, Discrete Mathematics Seminar, Texas State University, San Marcos, TX.

2016: Preliminary Results on Automated Conjecturing, Discrete Mathematics Seminar, Texas State University, San Marcos, TX.

2015:

- A Result Related to Vizing's Conjecture, Discrete Mathematics Seminar, Texas State University, San Marcos, TX.
- Definition Construction and Mathematical Inquiry (with S. Strickland), MAA Session on Teaching Inquiry, Joint Mathematics Meetings, San Antonio, TX.

2014: k -Power Domination and the k -Forcing Process, Computational and Applied Mathematics Graduate Seminar, Rice University, Houston, TX.

2013: The k -Residue of Disjoint Unions, CombinaTexas, University of Houston–Downtown, Houston, TX.

Departmental Service

Rice University.....

Spring 2024: Joshua Yaffee MIE Capstone Advisor.

University of Houston–Downtown.....

2019–2021: Chair of the Undergraduate Data Science Curriculum Committee.

2020–2021: Data Science & Statistics Tenure-Track Search Committee.

2017–2020: Mathematics and Statistics Colloquium Committee.

2018–2020: Mathematics Hospitality Committee.

2017–2019: Developmental Mathematics Committee.

Professional Service

2025–Present: Guest Editor, *Mathematics* Special Issue: New Perspectives of Graph Theory and Combinatorics.

2019–Present: Editorial Board Member, *Theory and Applications of Graphs*.

2019–Present: Referee for *Discrete Applied Mathematics*.

2019–Present: Referee for *Graphs and Combinatorics*.

2018: Referee for *Discrete Mathematics*.

2017: Referee for *Contributions to Discrete Mathematics*.

Teaching Experience

Rice University.....

Summer 2026: Data Science & Machine Learning (CMOR 438 / INDE 577). Graduate course cross-listed as an undergraduate course, Department of Computational Applied Mathematics & Operations Research.

Spring 2026: Data Science & Machine Learning (CMOR 438 / INDE 577). Graduate course cross-listed as an undergraduate course, Department of Computational Applied Mathematics & Operations Research.

Graph Theory (CMOR 404 / CMOR 504). Graduate and undergraduate course, Department of Computational Applied Mathematics & Operations Research.

Fall 2025: Data Science & Machine Learning (CMOR 438 / INDE 577). Graduate course cross-listed as an undergraduate course, Department of Computational Applied Mathematics & Operations Research.

Summer 2025: Data Science & Machine Learning (CMOR 438 / INDE 577). Graduate course cross-listed as an undergraduate course, Department of Computational Applied Mathematics & Operations Research.

Spring 2025: Data Science & Machine Learning (CMOR 438 / INDE 577). Graduate course cross-listed as an undergraduate course, Department of Computational Applied Mathematics & Operations Research.

Fall 2024: Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Spring 2024: Deep & Reinforcement Learning (INDE 597). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Fall 2023: Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Spring 2023: Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Fall 2022: Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Summer 2022: Deep & Reinforcement Learning (INDE 597). Masters of Industrial Engineering graduate course, Department of Computational Applied Mathematics & Operations Research.

Spring 2022: Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational and Applied Mathematics.

Fall 2021: Data Science & Machine Learning (INDE 577). Masters of Industrial Engineering graduate course, Department of Computational and Applied Mathematics.

[University of Houston–Downtown](#).....

Spring 2022: Statistical & Machine Learning (DATA 4319). Undergraduate course, Department of Mathematics and Statistics.

Calculus I (MATH 2401). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 2 sections.

Fall 2021: Statistical & Machine Learning (DATA 4319). Undergraduate course, Department of Mathematics and Statistics.

Introduction to Proofs (MATH 2409). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 2 sections.

Summer 2021: Calculus II (MATH 2402). Undergraduate course, Department of Mathematics and Statistics.

Linear Algebra & Multivariable Calculus (MATH 2422). Undergraduate course, Department of Mathematics and Statistics.

Spring 2021: Statistical & Machine Learning (DATA 4319). Undergraduate course, Department of Mathematics and Statistics.

Real Analysis I (MATH 3307). Undergraduate course, Department of Mathematics and Statistics.

Differential & Integral Calculus (MATH 2421). Undergraduate course, Department of Mathematics and Statistics.

Fall 2020: Statistical & Machine Learning (DATA 4319). Undergraduate course, Department of Mathematics and Statistics.

Introduction to Proofs (MATH 2409). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 3 sections.

Summer 2020: Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics.

Spring 2020: Statistical & Machine Learning (DATA 4319). Undergraduate course, Department of Mathematics and Statistics.

Differential & Integral Calculus (MATH 2421). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics.

Introduction to Computation with Python (CS 1311). Undergraduate course, Department of Computer Science & Engineering Technology.

Fall 2019: Introduction to Graph Theory (MATH 4308). Undergraduate course, Department of Mathematics and Statistics.

Operations Research (MATH/STAT 4311). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 2 sections.

Summer 2019: Statistical Analysis for Business Applications I (STAT 3309). Undergraduate course, Department of Mathematics and Statistics.

Spring 2019: Statistical & Machine Learning (DATA 4319). Undergraduate course, Department of Mathematics and Statistics.

Advanced Linear Algebra & Optimization (MATH 3423). Undergraduate course, Department of Mathematics and Statistics.

Differential & Integral Calculus (MATH 2421). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 2 sections.

Fall 2018: Differential & Integral Calculus (MATH 2421). Undergraduate course, Department of Mathematics and Statistics.

Introduction to Proofs (MATH 2409). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 3 sections.

Precalculus with Trigonometry (MATH 1505). Undergraduate course, Department of Mathematics and Statistics.

Summer 2018: Directed Study in Mathematics (MATH 4399). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics.

College Algebra (MATH 1301). Undergraduate course, Department of Mathematics and Statistics.

Spring 2018: Introduction to Proofs (MATH 2409). Undergraduate course, Department of Mathematics and Statistics.

Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 3 sections.

Fall 2017: Discrete Mathematical Structures (MATH 2305). Undergraduate course, Department of Mathematics and Statistics, 3 sections.

College Algebra (MATH 1301). Undergraduate course, Department of Mathematics and Statistics.

Intermediate College Algebra (MATH 1300). Undergraduate course, Department of Mathematics and Statistics.

[Texas State University](#).....

Summer 2017: Discrete Mathematics II (MATH 3398). Undergraduate course, Department of Mathematics.

Spring 2017: Discrete Mathematics I (MATH 2358). Undergraduate course, Department of Mathematics, 2 sections.

Pre-Calculus (MATH 2407). Undergraduate course, Department of Mathematics, 2 sections.

Fall 2016: Discrete Mathematics I (MATH 2358). Undergraduate course, Department of Mathematics, 2 sections.

Calculus (MATH 2417). Undergraduate course, Department of Mathematics, 2 sections.

Summer 2016: Discrete Mathematics II (MATH 3398). Undergraduate course, Department of Mathematics.

Spring 2016: Discrete Mathematics I (MATH 2358). Undergraduate course, Department of Mathematics, 2 sections.

Pre-Calculus (MATH 2407). Undergraduate course, Department of Mathematics, 2 sections.

Fall 2015: Math Bus & Eco II (MATH 1329). Undergraduate course, Department of Mathematics, 2 sections.

Math Bus & Eco I (MATH 1319). Undergraduate course, Department of Mathematics, 2 sections.

College Algebra (MATH 1315). Undergraduate course, Department of Mathematics, 2 sections.